

Dynamic Programming Problem Sheet

Fibonacci-Based DP and 0/1 Knapsack Problems

Part 1: DP on Fibonacci Concept

- 1 Classic Fibonacci number (nth term).
- 2 Number of ways to climb stairs with 1 or 2 steps at a time.
- 3 Tiling a $2 \times n$ board with 2×1 tiles.
- 4 Number of ways to climb stairs with steps {1, 2, 3}.
- 5 Minimum cost climbing stairs (LeetCode 746).
- 6 Count binary strings of length n with no consecutive 1s.
- 7 Decode Ways (LeetCode 91).
- 8 Space-optimized Fibonacci / stairs.
- 9 Matrix exponentiation to compute nth Fibonacci in $O(\log n)$.
- 10 Find nth Fibonacci mod $1e9+7$ for large n (up to $1e18$).
- 11 Ways to cover a distance with steps 1, 2, 4.
- 12 Number of compositions of n using {1, 3, 4}.
- 13 Frog jump with minimum cost (AtCoder DP A).
- 14 House Robber problem.
- 15 Number of ways to reach nth cell in a game board with dice rolls.

Part 2: 0/1 Knapsack and Variants

- 1 Classic 0/1 Knapsack — maximize value within weight capacity.
- 2 Subset Sum Problem — check if a subset sums to target.
- 3 Equal Sum Partition Problem.
- 4 Count the number of subsets with a given sum.
- 5 Minimum subset sum difference.
- 6 Target Sum (LeetCode 494).
- 7 Unbounded Knapsack (coin change type problems).
- 8 0/1 Knapsack with exact weight requirement.
- 9 Multiple constraints (weight + volume).
- 10 Knapsack with items having dependencies (tree knapsack).
- 11 Bounded Knapsack (limited copies of each item).
- 12 Knapsack with fractional items (Greedy) — comparison case.
- 13 Knapsack with maximum number of items constraint.
- 14 Bitmask DP knapsack variant.
- 15 Knapsack with special offers/discounts on items.